

# Real Time Visual Effects for the Technical Artist

Chris Roda

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# *REAL TIME VISUAL EFFECTS FOR THE TECHNICAL ARTIST*



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Chris Roda


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From a new teacher to my original teacher/mentors:

To Jeffery Light who lit the fire and encouraged me to start this VFX journey.

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# Author

**Chris Roda** is a Technical Art instructor at the Florida Interactive Entertainment Academy (FIEA), a graduate degree program in interactive, real-time application development at the University of Central Florida. Early in his career, Chris was a visual effects artist in the film and television industries where he contributed visual effects for films such as *Spider-Man, Titanic, and The Fifth Element*. Before coming to FIEA, Chris was a CG Supervisor at Electronic Arts, where he worked on video game titles such as *NCAA Football* and *Madden NFL Football*. In addition to teaching, Chris works on generating tools and pipelines for the creation of immersive experiences: the amalgamation of the narrative of films, the interactivity of video games, and the immersion of theme parks.



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# Introduction

Visual effects have been my passion for over 30 years. It has been my gateway for using computer graphics to generate alternative realities. Visual effects make the unbelievable real. They make miracles and the impossible happen. They create worlds, both in fantasy and in real life, that could not possibly exist. They also take what is very real but is imperceptible and make them visible. Visual effects invent fantastic and unbelievable elements that look and feel at home in whatever environments they are placed in. Most importantly, visual effects create story world credibility. They help cement participant presence in whatever game, film, or experience they collaborate with. When done correctly, visual effects provide the vehicle which gently immerses participants, without question, into game worlds or narrative experiences.

## WHY THIS BOOK?

Visual effects are an enormously vast and possibly complex realm of study. In the early days of off-line, film production, visual effects were often unexpected phenomena generated during the three-dimensional rendering process. Visual effects artists were often the best digital artists who understood the ins and outs of the entire rendering pipeline and were able to exploit its eccentricities to generate fantastic yet desirable results. They were able to extract components from each stage of computer graphics production, reconfigure, and re-combine them to create imagery that was never before possible.

As hardware and software improved over time, so did the visual effects. New and friendly interfaces make the progression for generating visual effects easier, more predictable, and more efficient. A visual effects artist no longer needs to understand the intricate details of the rendering process. Armed with traditional art training, a solid understanding of visual design and the right tools, an artist can create almost anything. Many artists pride themselves in not being technical as they surrender themselves to the purest creative expression.

Ironically, as the interfaces have improved and become friendlier, the underlying technology has continued to expand geometrically. In order to be made user friendly, the technology must be tamed, limited and bound sufficiently to be manipulated by the exposure of a minimal set of dialable parameters. The unlimited technological potential is still there, it is just hidden beneath amiable interface layers. A visual effects artist can be only as good as the exposed tools. All that is required to tap into that potential is a bit of understanding of the underlying systems and the willingness to crawl under the hood and get dirty with some oily math.

Metaphorically, this book is an introduction to hot-rodding real-time rendering engines. When a sports car is purchased from the dealer, the buyer possesses a very highly crafted piece of hi-tech machinery. Without any understanding of auto-mechanics, with only the control of the accelerator, brake, and steering wheel, the driver has a commanding presence on the road. However, when the driver wishes to tap into the true potential of the

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vehicle, a small amount of mechanical familiarity is required along with the willingness to make a few small modifications to the engine, exhaust, and suspension. Only then will the real potential of the vehicle be exposed. Real-time rendering engines are the same in spirit. With a little bit of understanding of the underlying technology and a rudimentary set of math skills, the visual effects artist can put the real-time engine into overdrive.

The contents of this book re-introduce real-time engine fundamentals in a language that is comprehensible by any willing artist. Only the basic concepts are discussed. All the gory details are left in the technological conference journals. However, when the artist wishes to further explore any of the topics presented in this book, sufficient nomenclature provides suitable signposts and directions for deeper exploration. This book is written from an artist's and a teacher's perspective in such a way that it will inform and educate but not drown the reader in mathematical complexity.

The world of visual effects is immeasurably vast. There are an infinite number of good solutions for any type of problem. The question is not so much how to solve the problem but what solution solves the problem best considering the context of the situation. With a subject area that is so broad, how can a single book be written about it? There may be no way to cover every specific topic in essential depth within one text. The best visual effects artists realize this early in their career and don't try to master everything. They reinvent prior solutions to fit in the context of new situations.

Realistically, visual effects cannot be mastered in one semester. They are a lifelong learning pursuit. This book does not try to be a definitive guide but attempts to introduce the broad topic areas that contribute to the realm of visual effects and demonstrates simple and common solutions addressing each topic area. Rough coverage of the fundamentals is provided so that any artist can modify any strategy to address any situation. This is not a book of tutorials. This book provides essential coverage of the fundamental tools visual effects artists should have in their tool kit so they can address any situation with courage as well as creativity.

## INTENDED AUDIENCE

This book is intended for artists wishing to understand visual effects topic areas and their fundamental solution strategies. It is written from an artist's perspective but is in no means dumbed down. There is math in this text. However, math does not venture beyond basic algebra and geometry levels. There is no specific code included. There are, however, strategies and algorithms provided that can easily be implemented within the context of any modern game engine using common and widely used tools and functions.

This text was originally written to supplement courses created for undergraduate and graduate-level game developers. It does not assume prior technical or coding experience. The techniques in this book can be implemented by experienced artists wishing to expand their existing palette of tools or for the programmer seeking an introduction to the impact visual effects has on the rendering pipeline. Producers and designers may also be interested in this book as an introduction to the language and terminology employed by all visual effects artists.

This is a book written for artists and technical artists. High technical, white-paper information will not be found here. The information is kept simple and abstract to be implemented in any real-time rendering engine or any digital content creation (DCC) software. This text is intended to be software agnostic. Visual effects can be juggled in just about any modeling/rendering/game engine software. Part of what makes visual effects artists' valuable is their ability to adapt and improvise based on their available resources.

There are no tutorials in this book. The information is presented theoretically but is absolutely implementable in any package. Tutorials tend to focus on implementing a single type of effect on one type of software. From this book's perspective, when the fundamental concepts are understood they can be integrated into any pipeline. When the core ideals are understood, they can be bent, warped, and melded with other ideals without being concerned with the interface idiosyncrasies of a particular software.

### Visual Effects Artists versus Technical Artists

There is an ongoing discussion concerning if visual effects artists are to be classified as their own independent tribe or as technical artists. This classification varies greatly between industries and production companies. This book attempts to treat both equally and interchangeably. Visual effects artists may identify themselves as digital artists who are focused exclusively on the art and design of the phenomena they are producing, using only the tools they are provided. Technical artists derive from a similar perspective but employ additional math and create new tools to complement the creative workflow. This book treats both as artists first. There is a considerable amount of technology introduced in this book but presented at a level non-technical artists can easily understand and discuss with technical artists the necessary strategies for future tool fabrication.

## HOW THIS BOOK IS ORGANIZED

This book is broken into seven chapters. The first three chapters introduce visual effects and establish fundamental topics all artists should be aware of. The last four chapters are devoted to visual effects categories and strategies used to implement them.

- **Chapter 1: What are Visual Effects?** explains the distinctions between special effects and visual effects and explores all possible realms which may be considered visual effects.
- **Chapter 2: Visual Effects Rules,** provides a collection of the best and most effective mental strategies a visual effects artist can adopt to ensure a rewarding and long-lasting career.
- **Chapter 3: Color Theory,** explains the usage of color from the perspective of a digital artist. Working with pixels instead of pigments has dramatic consequences that all digital artists need to be aware of in order to master the medium.
- **Chapter 4: In-Camera Effects,** explains the importance of lighting as the most profound and fundamental of all visual effects. A quick introduction to the tools and art of lighting is provided. This chapter also introduces the fundamental concepts of principle in-camera, otherwise known as post-process effects.
- **Chapter 5: Material Effects,** introduces the incredibly vast and powerful techniques of creating visual effects within objects' materials.
- **Chapter 6: Simulations,** covers the benefits and drawbacks of using simulation effects and introduces the foundational tools for creating and deploying simulations without destroying application framerate.
- **Chapter 7: Particles,** introduces particle effects as the amalgamation of lighting and in-camera, in-material and simulation effects. The chapter covers the composition of particles, how they are integrated with the other types of effects and general strategies for their rendering.
- **Chapter 8: Conclusion,** concludes this book with the classifications of visual effects which do not necessarily fit within the designations of earlier effects, where the visual effects industry is headed in the near future, and a basic strategy for starting a career in visual effects.

## CONVENTIONS

Many of the terms and applications mentioned throughout this book may seem to go in multiple directions and even be contradictory. To avoid confusion, the conventions used throughout this book are listed below.

### Experiences

Without getting too philosophical, real-time visual effects have more applications than just video games. All of the effects presented in this book may have originated from games but are now usable in any real-time application spanning from games to simulation and education, architecture, medicine, marketing, industrial and any form of virtual, augmented, or mixed reality experiences. These are momentous times as even traditional narrative media such as film, television and live events are transitioning toward real time. Instead of listing all possible avenues upon which visual effects can be applied, this book simply refers to all possible applications as, "real-time experiences," or simply, "experiences."

### Story Worlds

Just as all real-time experiences are no longer the domain of just video games, the thematic environments which visual effects are surrounded by and attempt to integrate with must be re-labeled. The convention employed by this book is "story worlds." The use of this terminology is not intended as a slight against games which may be more focused on gameplay over story. The term story world describes the thematic environment which surrounds the real-time application. Even gameplay driven games such as Tetris and Pacman take place within some type of story world pretense, even as simple as, "An electronic maze filled with hovering pellets which are consumed by a munching disk while being pursued by ghosts in the machine."

### Art Directable

Throughout this book, there is continuous reference to an adverb, "art directable." Art directable refers to the ease at which the visual content of the visual effect is changed or updated, based on recommended feedback. For example, when observing a visual effect, an art director may advise the effects artist to increase the fire intensity of an explosion, reduce the smoke, increase the sparks and make the upper surface of the exploded object dissolve more rapidly than the lower portions. Increasing the number of sparks may be very art directable and simple to implement. Depending how the effects artists implemented the effect, changing the distribution of the dissolving object may not be art directable and could require an entirely different design to the effect.

### Users

From this book's perspective, there are no differences between Visual Effects Artists and Technical Artists. They both produce real-time visual effects. Visual Effects artists may focus more on the art and design aspects of the phenomena they are producing. Technical Artists deliver the same results but may employ more mathematics or invent their own tools to aid in delivery. Regardless of the tools they employ for creation, both produce similar results and therefore their labels are interchangeable.

## SOFTWARE DISCLAIMER

This book attempts to be software agnostic. All presented strategies are universal and should be applicable to any real-time rendering engine. All digital content creation tools (DCCs) used for generating the assets populating these engines are intended to be universal as well. Great effort was made to avoid reference to any software

specific applications or features. For consistency, all of the images generated for this book were produced using Epic Unreal Engine, SideFX Houdini, and Adobe Photoshop. This book is not an endorsement for any of these products. They just happen to be the author's favorite tools of choice. Any real-time engine should be able to reproduce these results. Any DCC should be able to generate assets to use in these engines.

## PREREQUISITES

This probably comes across as cliché as all texts request this of their readers. Not much is needed in order to explore this book's contents other than a willingness to learn. Many artists, even self-proclaimed technical artists, declare they don't do math. Maybe they had a bad math-related experience in middle school or had never been presented with a useful application for the math they were learning. Others proclaim they don't deal with technology as it interrupts the creative process. This is always surprising since they are practicing digital art in a technical medium.

Regardless of the barriers, the greatest artists have always been masters of their media. Great painters understand enough chemistry and physics to mix their own pigments, fashion their own brushes, and stretch their canvases. Sculptors blend their own substances and fabricate their own tools. Luckily, visual effects only require a small amount of physics. With a small amount of math and technical understanding, visual effects artists can increase their potential for artistic expression exponentially.

This book requires basic addition and subtraction complemented with some multiplication and division. Technical artists traditionally only use multiplication. They use numbers larger than one to increase a product, smaller than one to make it smaller and negative numbers to represent the opposite or flip directions. Small amounts of algebra and basic geometry are useful for solving formulas. Calculus and linear algebra are great disciplines and will significantly help visual effects artists in the long run. However, they are not essential for this book nor are they requirements for entrance into the field.

A real-time visual effects artist must have a real-time rendering engine to practice her skills. Unreal Engine and Unity, at the time of this book's writing, are very popular engines. Their basic functionalities are free to use and don't require payment until certain monetary thresholds are achieved using those engines. Aside from Unreal Engine and Unity, there are many other engines which are free to use, such as Lumberyard from Amazon. Many engines, such as Godot, are open-source. While most larger studios use customized engines, the availability of free-to-use engines should not slow the visual effects artist.

The choices for DCC software used for generating in-engine content are as diverse as real-time engines. While there are expensive three-dimensional creative suites available, such as Side Effects Houdini, Autodesk Maya, and 3DS Max, each should provide a free trial period or free learner's version in order to fine tune one's skill. Blender is an upcoming open-source, three-dimensional package. Two-dimensional image manipulation software is also required. Commercial packages such Adobe Photoshop and Illustrator are industry stands. Open-source and free-to-use tools such as Gimp and Krita are also excellent packages.

There has never been a better time to start one's journey into real-time visual effects. The availability of outstanding real-time rendering engines and DCCs has removed most barriers from entering the field. There is a plethora of excellent material on-line to demonstrate almost any technique expected from a starting visual effects artist. A visual effects artist should have no limitations for exploring this diverse field.



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## INTRODUCTION

When considering the components that make up the real-time production process, visual effects are the most difficult to define. For brevity, instead of defining all that visual effects are, it may be easier to identify what visual effects are not. Within any computer-generated scene, the participant will find multiple familiar items: environments, animated characters, and props. Artificial environments define the story world's settings, bounding what is conventional. They introduce the participant to what is typically expected or what is plausible for the scene. Characters are animated from keyframes or from motion capture. They interact with the environment and other characters. Props are passive objects with which both characters and environments act upon. Visual effects are everything else found in that story world. When a component is not considered an environment piece, an animated character, or prop, it is a visual effect (Figure 1.1).

Visual effects are the lighting that make a forest look foreboding in the middle of the night and enthusiastic in the morning. They provide the camera flares in the early sun and make the world appear blurry in times of obscurity and confusion. Visual effects are what make static objects appear as if they are moving, such as barber poles or wind-blown foliage, and make water appear to be flowing and rippling. They generate dramatic destruction such as collapsing buildings and the subtle bounce of a princess's hair as she rides her horse. When not captured as on-set special effects, visual effects provide rain, snow, smoke, fire, and rainbow-colored pixie dust from magical wands.

Generating emotional impact is a primary objective of the computer-generated pipeline. Skillful writing and compelling performances drive computer-generated imagery (CGI) characters to interact with their environments and props to create memorable visual experiences. Visual effects complement these fundamental components and influence emotional mood and tone. They make dire situations feel more dreadful and happy situations more

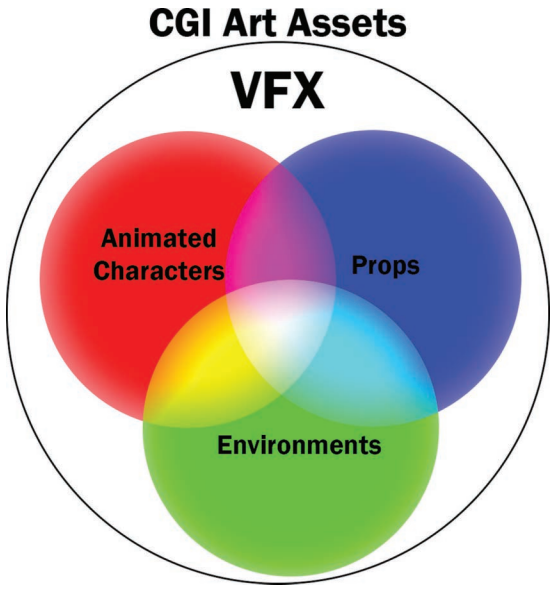



**FIGURE 1.1** Visual effects in the computer-generated scene.

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festive and alive. They make mundane events feel like epic, destiny threatening circumstances. Visual effects complement and supercharge scene components to create emotionally driven adventure.

Real-time environments can be anything from the mundane to the outrageously unbelievable. They may represent any moment in time from the present to the ancient past and to the unforeseeable future. Environments may take place underwater, within outer space, or in alien worlds. Visual effects make these settings feel real, natural, and believable. They add credibility to their surroundings making them feel as if they have always existed.

Story world credibility is threatened when alien beings or fantastic creatures are introduced. Foreign characters are thematically integrated with familiar settings through the exchange of visual effects techniques. For example, consider a frog demon summoned by an evil wizard to wreak havoc on a medieval hamlet. Simply introducing the character into the environment without the aid of visual effects feels artificial and cheap. Adding visual effects makes the demon creature, although very alien, feel natural within the experience. The lights of the village illuminate the demon while the fire in its eyes illuminates itself as well as its immediate surroundings. The lighting in the town casts shadows onto the demon and conversely, the demon casts shadows back onto the environment. When the demon collides with scene objects, they are damaged or destroyed. When the demon lumbers by, it leaves footprints kicking up dirt and sand. The creature spits slime and soaks the surroundings and all who come within immediate contact. The exchange of visual effects integrates alien components within familiar environments.

Visual effects take an ordinary experience and make it larger than life. They push the experience from the real to the hyper-real. Very rarely do the events of real-life match the emotional impact generated by the same situations. Instead of displaying what is real, visual effects create the story world constructed by our emotions. Visual effects push the impact of natural phenomena to the catastrophic. They add vibrant colors and brilliant lighting to otherwise mundane situations. Visual effects make reality conform to the emotional stories created with our imaginations.

From a pure commercial perspective, visual effects are the components added to an experience to cue the audience something special or wonderful is occurring. This is the extra icing on the cake, the extra polish to the scene, or the extra glitter to the gold. Visual effects appeal to the curiosity and invite outsiders to participate within the offered story world.

Visual effects are diverse. They can be anything from the lighting that controls the scene's emotional mood and tone to the shapes a frog morphs while transforming to a prince. They can be the dark clouds that roll on to a field of battle to the armies that lob projectiles at each other. Visual effects can be the blood and destruction spilt on the battlefield as well as the foliage that grows and replaces them. Instead of replacing characters, props, environments, and their animations, visual effects complement these components within their thematic story worlds.

## VISUAL EFFECTS ARE ANIMATIONS

All visual effects are animations. The visual effects industry evolved from animation. All original visual effects artists were former animators or had mastery of the 12 principles of animation. Visual effect images need to move and change over time; otherwise they are simply pretty pictures. This condition even covers lighting and post-processing which may not appear to change. Over the course of an experience, they can and often do change. The contrast of the same scene between two different contexts has a dramatic impact on the experience's emotional mood and tone.

## SPECIAL EFFECTS

There is an important distinction to be made between visual effects and special effects. While visual effects are important components of the CGI production pipeline, "Special Effects" is a live-action production term reserved for phenomena requiring unique attention. Usually mechanical in nature, these phenomena may include mechanized props and scenery, scale models, animatronics, pyrotechnics, and atmospheric simulations such as physical wind, rain, and snow. Examples of mechanized props are exploding toilets, collapsing stages, and imploding buildings.

Special effects also include prosthetic makeup that makes actors look non-human. These include any mask or prosthetic that moves or behaves in an unnatural way.

Before the evolution of computer visual effects, all optical effects were considered as special effects. These phenomena were manipulated after primary image capture and were considered as part of the "in-camera" process. Source images were transferred from source film to destination film using optical printers, mattes, and other multiple exposure techniques.

Computer-generated imagery originated as a subsection within special effects and evolved into its own area of specialization. Now, any imagery created or manipulated in post-process using a computer is considered as a visual effect while any phenomena captured concurrently with live footage is considered a special effect.

## OFF-LINE VISUAL EFFECTS

Any special effect captured or manipulated after primary exposure and before final coloration was considered as part of the "in-camera" process. These were also called "off-line" effects because the techniques were performed away from the expensive on-set production process. These effects included optical effects, models, and miniatures which were filmed and optically integrated with source footage. Being off-line meant extra time and attention could be devoted to the special effects without impacting or delaying primary production.

Computers were originally used to assist with the off-line filming of models and miniatures. Computer motion-controlled camera rigs generated impossibly complex camera movements and multiple camera passes which duplicated camera movements with every iteration. Different camera passes were filmed under different lighting conditions to generate mattes and isolate diffuse, shadow, and highlight information layers. Computers were also used for assembling these layers during optical printing.

Three-dimensional CGI was first used in the 1976 film, *Futureworld*. Used sparingly afterward, the use of CGI gained significant attention in the 1982 films *Tron* and *Star Trek II: The Wrath of Khan*. Other landmark films such as *The Last Starfighter*, *Young Sherlock Holmes*, *The Abyss*, *Terminator 2: Judgment Day*, and *Jurassic Park* contributed CGI to evolving into a significant special effect technique. CGI continued to replace optical special effects until evolving its own *visual effects* category.

The 1995 film *Toy Story* became the first full CGI film. Instead of providing only special effects components, computers were used for generating all visual production content. This landmark event initiated a transition of many animated films from hand-drawn to computer generated. While computers are still used for generating CGI films, the process is still relatively slow. This limitation is changing radically.

In this book, the term *off-line* refers to all CGI production performed after original source exposure and before final coloration. While this term applies to all live-action visual effects, it also describes the production process for CGI-animated films. Being off-line frees artists from generating frame-rate dependent images, allowing focus on quality over performance (Figure 1.2).

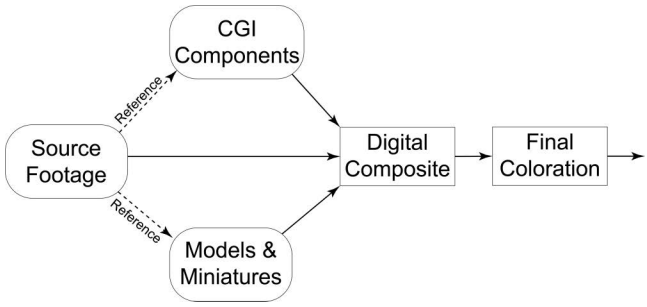



**FIGURE 1.2** Off-line production process.

## REAL-TIME VISUAL EFFECTS

Real-time visual effects are components of larger computer applications running fast enough to display fully rendered images at frame rates between 30 and 90 frames per second. Compromises to the visual effects must be made to prevent them from being computationally intensive enough to impact the framerates of their larger programs. Off-line visual effects are not impacted by this condition.

Although considered a relatively modern technological development, real-time computer graphics are almost as old as off-line. In the mid-1980s, the rise of Silicon Graphics and their three-dimensional computer graphics workstations empowered the first commercially available, real-time, three-dimensional simulations. The image quality was limited and primitive but did not prevent Silicon Graphics from generating a presence in the simulation and visualization industries. During the mid to late 2000s, high-end graphics workstations became obsolete as three-dimensional graphics cards became available. Graphics cards or Graphics Processing Units (GPUs) have continued to become faster and more powerful. As of the time of this book's writing, GPUs are now capable of real-time, ray-traced rendering; something that was considered pure science fiction in the 1980s. GPUs render the same three-dimensional worlds in real-time that once took weeks of off-line processing time in the 1990s. The distinction in quality between off-line and real-time rendering is now blurring.

In many considerations, real-time visual effects are more difficult to create than off-line. While their level of visual quality may not always be as sophisticated (although that line is blurring), the real-time condition introduces added complexities. The relatively free-roaming, real-time camera prevents hiding information from pre-established camera views. Because the content creator cannot control the direction of participant focus, all scene elements must appear consistent from all camera angles. Thematically, off-line production can manipulate audience attention to maximize narrative impact. This ability is greatly reduced in real-time and disappears entirely with augmented and virtual reality applications. What was once occluded in off-line scene composition is now exposed in real-time. This exposure requires greater project pre-planning and resources to ensure visual consistency from all camera perspectives. Effective real-time visual effects, now more than ever, must be sensitive to thematic narrative.

## DRAWBACKS OF REAL-TIME VISUAL EFFECTS

Although the quality lines separating off-line from real-time visual effects are indistinguishable, real-time visual effects have a significant drawback. All visual effects must be authored within the real-time rendering engine. This phenomenon is explained more thoroughly later in this book. No two rendering engines are the same and therefore cannot produce the exact same visual results given the same input. There are many outstanding, off-line DCC packages that are effective for generating off-line visual effects. The output from these packages do not translate directly into mass-market real-time rendering engines. The data must be conditioned and re-configured to be input into real-time engines. When rendered, the differences between the off-line and real-time renderers prevent the two from maintaining visual parity.

Just because off-line and real-time packages cannot duplicate each other's appearance, does not mean that they cannot support each other. Off-line packages provide outstanding pre-visualization and prototyping services which the real-time packages use as visual targets. Techniques such as "Vertex Caching" enable off-line software to pre-generate sophisticated simulations that can be imported into real-time. These "pre-baked" simulations, when played back in real-time, are exponentially more sophisticated and more efficient than any simulation generated in real-time.

## REAL-TIME USES

What started with Silicon Graphics in the mid-1980s with simulation and scientific visualization has snowballed into a diverse range of applications. Real-time applications now reach into every aspect of multimedia. Visual effects complement every application.

### Video Games

Video games are the first applications that come to mind when considering real-time visual effects. From atmospherics such as rain and snow, to simulations depicting epic destruction and to the visualizations of the unbelievably fantastic, visual effects are essential components for generating immersive and captivating gaming experiences (Figure 1.3).

### Live-Action Production

Live-action production, which was once strictly off-line, is now transitioning to embrace real-time production methods. Pre-visualization empowers directors to observe and interactively compose thematic scenes before sets are built and performances are captured, saving productions phenomenal cost, and guaranteeing effective narrative intent. Set design and lighting are pre-modeled to provide virtual blackboards to work out lighting and other production challenges before live-action capture. Integrated with traditional green-screen techniques and virtual environments, real-time composting provides acting talent holistic environments to interact with and avoids further post-production visual effects (Figure 1.4).



**FIGURE 1.3** League of legends magic. (Image Sources Courtesy of Riot Games.)



**FIGURE 1.4** UE4 virtual production. (Image Courtesy of Epic Games © 2021, Epic Games, Inc.)

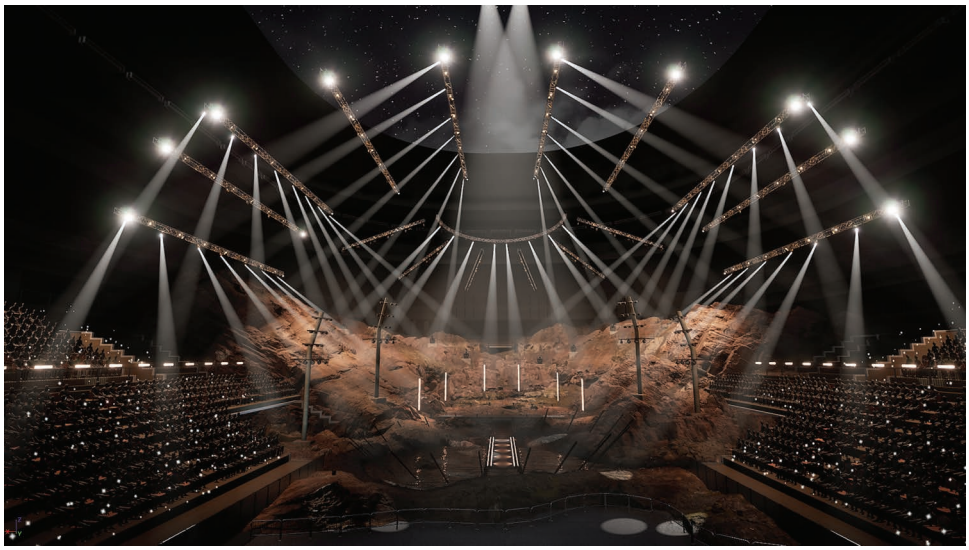


**FIGURE 1.5** Live-action sets and visual effects. (Image Courtesy of Epic Games © 2021, Epic Games, Inc.)

Live entertainment production has also integrated with real-time. Concerts, competitions, sporting events, and other live-action events are pre-visualized and pre-lit in real-time. Set extensions and environmental control are pre-choreographed with stylish motion graphics to produce impactful and memorable live experiences (Figure 1.5).

## Cinematics

Cinematics are real-time short movies. Performers and their environments are generated and played back entirely within real-time. Cinematics are common narrative components found in video games. They are now found in marketing and episodic venues. Full-length cinematic productions are not far into the future (Figure 1.6).



**FIGURE 1.6** Full-length cinematic venture. (Image Courtesy of Spire Animation Studios.)

### Simulation

Simulation continues to grow as a significant real-time application. Visualization, training, and modeling industries are dependent on real-time content for aiding doctors, soldiers, healthcare providers, and first responders by providing safe and consequent-free environments to develop and fine-tune lifesaving skills. Visual effects are crucial components for establishing realistic training scenarios.

### Augmented Reality (AR), Virtual Reality (VR), Mixed Reality (MR), and Cross Reality (XR)

While immersive virtual reality has been around almost as long as real-time CGI, the development of free-to-use real-time engines and consumer-focused head-mounted displays has made these elements to of our everyday lives. Augmented and virtual reality applications are dependent on real-time display. Visual effects are essential components for integrating and establishing alternative reality integrity.

### Architecture

Architects have been using three-dimensional computer graphics since the mid-1980s. Real-time environments empower architects and designers to rapidly iterate and produce the most effective versions of their creative visions and communicate their intent to stakeholders. They also help identify and navigate around costly design errors. Architects employ walk-though, immersive VR, and AR experiences to holistically juxtapose the past, present, and future (Figure 1.*7*).

### Automotive/Transportation

Real-time visualization has become a significant tool in the automotive and transportation industries. Like architecture, real-time solutions provide designers an environment to prototype and refine concepts and then interact with their results without having to build expensive physical models. While visual effects do not significantly contribute to this area, they are used extensively in the visualization and marketing departments. Visual effects bring products to life and guide audience attention.

Inside the driver's seat (or in the cockpit), the driving experience is pre-visualized in real-time so designers may understand the impact of the interior design for safe and effective navigation. Visual effects are crucial for



**FIGURE 1.7** Architectural visualization. (Image Courtesy of Imerza.)




**FIGURE 1.8** Real-time user interfaces. (Image Courtesy of Epic Games © 2021, Epic Games, Inc.)

prototyping realistic driving experiences. They are even used to create user interfaces which are easy to read and interact with (Figure 1.8).

Visual effects also aid in teaching vehicles how to drive by themselves. Games and simulations are created to teach autonomous vehicles how to safely operate without human interaction. Visual effects are integral components of these simulations.

### Scientific Visualization

An interactive, scientific visualization is almost always generated as a real-time visual effect. Visual effects techniques are essential for displaying natural phenomena from the microscopic worlds of viruses to the movement

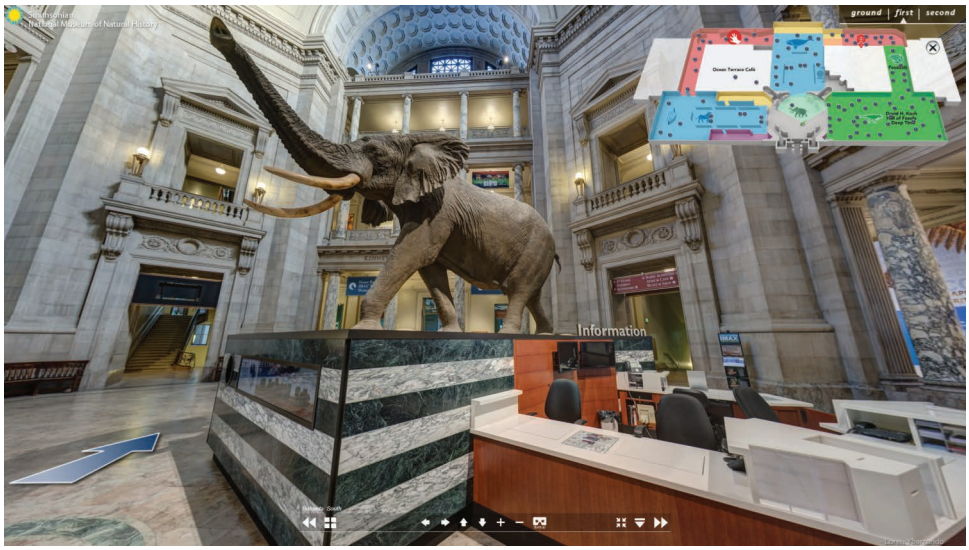



**FIGURE 1.9** Interactive cultural experiences. (Image Courtesy of the Smithsonian Institution.)

and behavior of galaxies. What were once exclusively computed off-line, real-time visual effects are core components for interactive scientific visualization.

### Location-Based Experiences

Real-time technologies dramatically impact museums, art galleries, and other location-based experiences. Interactive historical and cultural exhibits educate and inform while integrating the past, present, and future into singular contexts. Visual effects play an important role collaborating in real-time to emphasize content and generate larger-than-life experiences (Figure 1.9).

## REAL-TIME VISUAL EFFECT TYPES

Real-time demands make visual effects one of the most challenging of all technical art disciplines. They are the juxtaposition of the most complicated technical and the most creative design components. The first visual effects artists were the best of all digital artists. They possessed a solid foundation in art and design all the while manipulating and exploiting every aspect of the rendering pipeline. They needed to be proficient with the 12 principles of animation and write code to duplicate each. They were required to model, draw, and paint sufficiently to create magical characters and bring them to life. Due to the primitive hardware of the time, visual effects artists also understood system IT in order to drive the best computational performances. Visual effects artists were the first CGI Jacks of the trade.

With a seemingly endless variety of real-time applications, it is easy to conclude there is an infinite number of types of visual effects. To help wrangle this endless herd of kittens, this book breaks real-time visual effects into four categories based on implementation: in-camera, in-material, simulation, and particles. Of course, other categories might be considered as visual effects such as procedural modeling and procedural animation. However, examples such as these may be considered as technical extensions of traditional artists' roles and not typical visual effects situations. In this book, coverage for each visual effect type is limited to introducing each's fundamental processes and ideologies. The entire book could be devoted to each category implementation. Most modern visual effects are complex aggregates of simpler instances from each category. These categories are adequate to help any artist handle any visual effects demand.

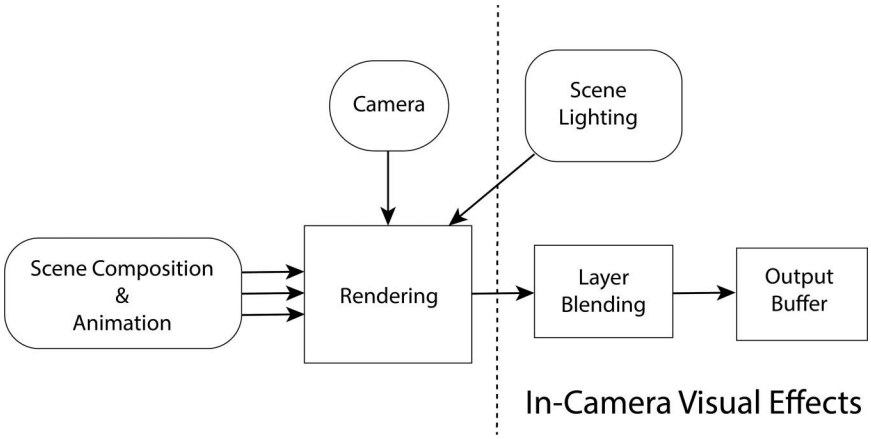



**FIGURE 1.10** Real-time in-camera visual effects.

### In-Camera Effects

In-camera visual effects are the closest in spirit to the original special effects employed in live-action production. While traditional off-line visual effects are everything done to captured images after initial exposure and before final coloration, real-time in-camera visual effects are all effects performed during or after initial scene rendering and before the final output-merger stage of the rendering pipeline. No additional visual effects actors or components are introduced to the pipeline (Figure 1.10).

These visual effects are broken into two sub-categories: lighting and post-processing. While not traditionally viewed as a visual effect, lighting, in the author's perspective, is the most important of all visual effects. Nothing in the art production pipeline has a greater impact on the emotional mood and feeling of the real-time experience than lighting (Figure 1.11).

Post-processing is all additional operations done to rendered image layers before the final merge. Since these manipulations are performed after initial rendering, post-processing significantly impacts overall experiential mood and atmosphere with the least amount of effort (Figure 1.12).

### Material Effects

Beyond in-camera effects, in-material effects are the most widely implemented type of visual effect. In-material effects are processes performed inside scene object materials which manipulate the stages of the rendering pipeline. Objects and characters with visual effect materials are rendered in separate layers from other scene components. All object materials are potential visual effect generators. Significant advantages of these types of visual effects are that they are relatively simple to implement, they are extraordinarily art directable, and when performed consciously, they do not draw heavily on computational resources. While this technique offers the greatest potential for variation, there are two primary strategies: animated texture maps and vertex manipulation (Figure 1.13).

### Simulations

While in-material effects are relatively computationally inexpensive, simulations are the most expensive. Simulations are movements and deformations computed in real-time to mimic natural physical phenomena. Because they duplicate the physics of experience's theme world, simulated objects integrate with their environments seamlessly. Simulations help establish the integrity and realism of the experiential universe. Instead of manipulating visual effects objects from within their materials, simulations animate and deform in world-space,



**FIGURE 1.11** Same scene lit in two different ways (bright and creepy).

from the context of the entire rendering engine, not the individual object. Simulations provide the most natural looking animations but are expensive and difficult to art direct (Figure 1.14).

### Particles

When most people think of visual effects, they think about particles, procedural simulations of tiny, abstract objects. Particles are the aggregate of all the other types of visual effects. They can manipulate lighting and provide layers of post-processing. They must be rendered and can take advantage of all the techniques available from the in-material rendering pipeline. They are also primitive simulations which reflect the physics of the story world's universe. From a pure programming perspective, they are difficult to implement, requiring specialized tools to create and edit. Ironically, because the tools and interfaces have evolved since the late 1980s, they can be implemented easily. They can take moments to learn yet require a lifetime to master. Their ease of entry makes them the most design intensive and art directable aspect of visual effects. As the aggregate of all other visual effect types, they are also the most complicated and technically demanding. All visual effects artists sit somewhere between these two radical extremes (Figure 1.15).

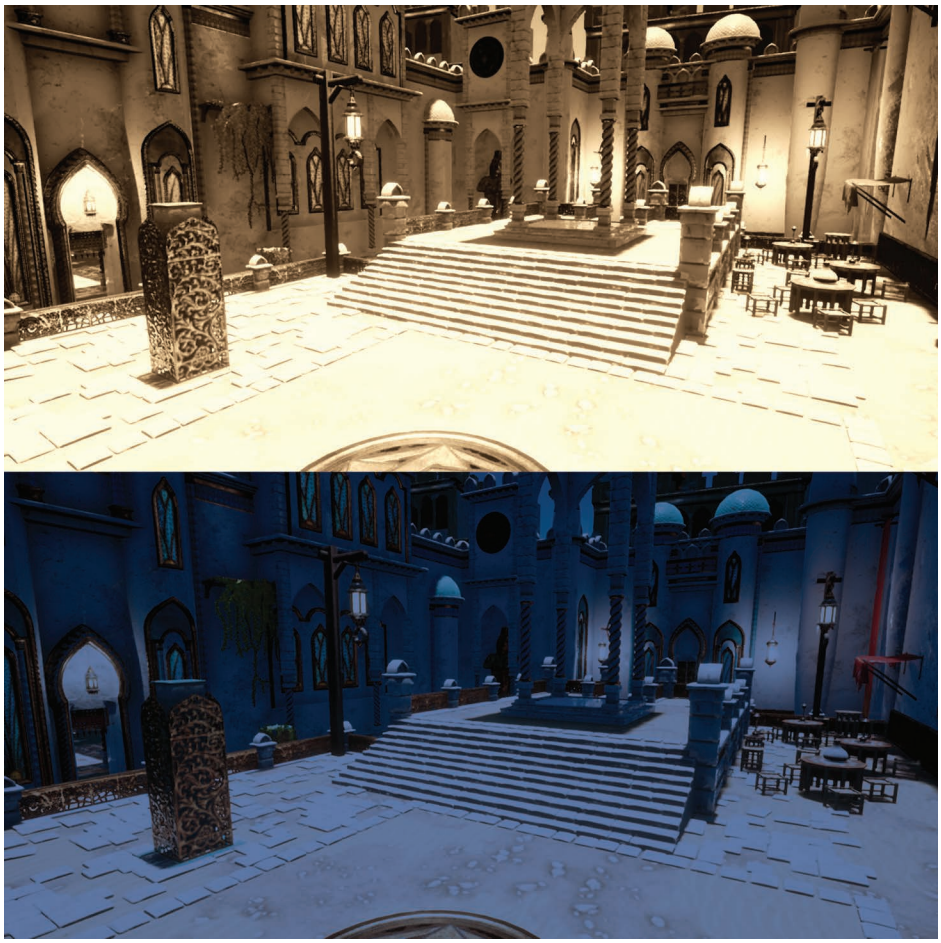


**FIGURE 1.12** Same scene with two different post-processes.



**FIGURE 1.13** In-material visual effects.



**FIGURE 1.14** Simulations.



**FIGURE 1.15** Particle visual effects.

## CONCLUSIONS

Real-time visual effects are the evolution of live-action special effects applied within real-time rendering engines. Special effects manipulate scene objects during live-action capture. Off-line visual effects manipulate exposed footage after initial capture and before final coloration. The off-line visual effects process is independent of frame rate and can be performed very quickly or very slowly. Real-time visual effects are framerate dependent and must not negatively impact the performance of the real-time rendering engine. Within the real-time art production pipeline, visual effects are all components which are not key-frame animated environments, articulated characters, or props.

Real-time visual effects complement and manipulate the mood and emotion of their story worlds, yet they are bound directly to their rendering engine. They must be visually mastered within the context of their rendering engine, making them difficult to implement and refine.

Despite their limitations, real-time visual effects are found in all real-time applications including video games, live-action production, cinematics, simulations, all forms of mixed reality, architecture, automotive and transportation, scientific visualization, and location-based experiences.

While infinite in possibilities, this book breaks real-time visual effects into four categories: in-camera, in-material, simulations, and particles.

## INTRODUCTION

Visual effects are the most creative aspect of being a Technical Artist. They provide an opportunity to focus on creating "fine art" and design. Creatively, the artist does not need to be technically savvy to create beautiful content. Ironically, visual effects are also technical enough such that the artist may focus purely on their science and engineering. Visual effects are a perfect juxtaposition of the artistic skills and the technical elements. Each artist controls the ratio of the two. This chapter focuses on the rules for creating and designing visual effects regardless of their technical or creative requirements.

Many of these rules are subjective. They may not be so much as rules as they are guidelines for creating pleasing visual effects. Some are based on technical conditions which carry subconscious impact, such as white and black points. These rules are ordered to provide logical flow that is easy to remember. Some may seem contradictory. These are situationally conditional rules that require the creative eye of the artist whose job is to balance the artistic skills and the technical elements.

This list of rules is not complete. It is continuously growing and expanding. The list refines itself as time and technology progress. The opportunity for integrating other artists' rules is exciting. The path for technical artist growth is not paved with ability but growth and learning. Any offered critique from this chapter is not intended to establish superiority but to provide opportunity to learn and understand from an alternative perspective. Analyzing any visual effect provides opportunity for growth.

Many of the concepts regarding artistic principles, narrative support, composition, and interpretation are derived from working in film and television production. These concepts are integrated with the constraints and technological demands of the real-time realm. Up and coming media formats such as virtual reality, augmented reality and mixed reality, integrate narrative and technical elements. Regardless of their medium, these rules guide the visual effects artist to creating more impactful and aesthetically pleasing visual effects.

## VISUAL EFFECTS ARE NOT THE STORY

Visual effects are fun. They are showy and spectacular. They draw attention to themselves. They are also subtle and may never be noticed. Regardless of their intention, visual effects are not story or the gameplay, they complement both. Unless the story is about the visual effects, they must support the story, not compete with it. When visual effects and story exchange priority, the experience feels artificial.

### Visual Effects Add Credibility

Visual effects make the unbelievable visible. They make impossible story worlds real. None of the magical game universes of *Skyrim* and *Azeroth*, the alien landscapes of *Star Wars*, or the dinosaurs of *Jurassic Park* would be possible without visual effects. Even when impossible in our own mundane reality, visual effects provide what is expected to be seen in the context of their story worlds. They provide what is necessary for the audience to believe the world has always existed. They provide credibility to the justification of alternative realities.

### Visual Effects Are the Glue

In a story world or gameplay universe, there are characters and environments which must be implemented as visual effects. Many characters are pure digital effects such as Thanos from the *Avengers* or a young Benjamin Button. The environments of *Rapture* and *Night City* are entire worlds composed of dizzying layers of visual

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